Wang Zhao



Software Engineer · Singapore · Singapore Citizen

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Two years on a large React and TypeScript micro-frontend platform in the defence sector, plus backend and deployment work.

EXPERIENCE

Software Engineer, NCS Jul 2024 – Present

Logistics platform for a government defence client, built as React micro-frontends over Spring Boot services.

- Build features for three of the platform's micro-frontend apps in React and TypeScript (Redux Toolkit, Ant Design, Module Federation), and fix and extend components in the team's shared Storybook library.
- Built saved table layouts, where each user's column visibility and order persist to the backend. The team later made it a standard customisation in the design system.
- Worked on embedded Qlik analytics dashboards and on shared features across the platform: help centre, notification centre, landing pages and account management.
- Restructured a Spring Boot file-validation service for a second, more conditional rule set: template-method and factory classes so processing steps can be reordered, and strategy classes for building rules.
- Sped up validation of large multi-file sets with parallel batch reads, and by computing each file's SHA while reading it line by line, so every file is read once instead of twice.
- Set up on-premise servers, wrote the setup and deployment guides, and deployed and debugged on client sites.
- Use the client's approved on-premise AI coding tools for tests and boilerplate, and maintain the repository's agent rules (for example, rules that block circular dependencies).

Software Engineer Intern, NCS May – Aug 2023

Wrote JUnit and Mockito tests for Spring Boot modules ahead of a migration to a new architecture.

Software Engineer Intern, Continental Automotive Singapore Jan – Apr 2023

Built an Angular dashboard for monitoring bus-driver health and performance, with engineers in Germany.

PROJECTS

Spendulu React, TypeScript, Vite, Pyodide, Web Workers · built with Claude Code

Personal-finance web app for Singapore, where banks offer no consumer API. Reads PDF statements from five local issuers and runs the Python parsing engine in a Web Worker, so statements never leave the device.

browhen Python, SQLAlchemy, Alembic, Postgres, Cloud Run, GitHub Actions · built with Claude Code

Telegram bot that finds a day a group chat can meet. The database enforces one open plan per chat, tests check the models against the migrations, and CI runs on every push.

LaparoLymph Unity, C#, VR

Final-year VR surgery trainer with Duke-NUS and Singapore General Hospital. Built all the software in a team of seven: anatomy viewer, operating-theatre scene, tool physics, guided procedure and scoring. Demoed to surgeons.

EDUCATION

Singapore University of Technology and Design, B.Eng, Computer Science and Design 2020 – 2024

NUS-ISS, Graduate Certificate, Designing Modern Software Systems 2025

Professional Scrum Developer I (PSD I)

SKILLS

Frontend: React, TypeScript, JavaScript, HTML/CSS, Redux Toolkit, Ant Design, Storybook, Module Federation, Qlik (Nebula.js), Angular
Backend: Java, Spring Boot, Python, Ruby on Rails, SQL, REST APIs, Microservices, Keycloak/OIDC
Testing: Jest, React Testing Library, JUnit, Mockito
Tools: Git, CI/CD, Agile/Scrum, Jira, Unity, C#
Languages: English, Mandarin **Interests:** Bouldering, diving, snowboarding, dance